The strong DF-p-log-Sobolev range for nontrivially decohering QMS

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Source. This packet concerns the open-question paragraph in Bardet–Rouze, arXiv:1803.05379, page 37. The source asks for the range of exponents p for which a non-primitive quantum Markov semigroup can satisfy a strong decoherence-free logarithmic Sobolev inequality

$$\mathrm{LSI}_{p,\mathcal{N}}(c,0)$$

for some finite $c > 0$. The paper proves a no-go theorem at $p = 2$, while the modified $p = 1$ endpoint is known to behave differently in examples.

> this raises the question of finding the range of p's for which one can find a non-primitive semigroup for which $\mathrm{LSI}_{p,\mathcal{N}}(c,0)$ holds for some $c > 0$. Moreover, such a no-go result implies the impossibility for any primitive QMS to satisfy $\mathrm{LSI}_{2,\mathrm{CB}}(c,0)$, as opposed to $\mathrm{LSI}_{1,\mathrm{CB}}(c,0)$.
>
> All these results rely heavily on the structure and the properties of the amalgamated $\mathbb{L}_p$ spaces. Further development will require better understanding of these spaces, in particular as interpolating Banach spaces.

Status. This packet gives a claimed full answer, subject to human review, for finite-dimensional nontrivially decohering QMS. It proves that no finite $p > 1$ admits a strong inequality. The $p = 1$ endpoint is the modified logarithmic Sobolev inequality considered in Bardet's earlier work, where positive examples are known under standard regularity hypotheses. Thus the existence range is exactly the endpoint $p = 1$.

Definitions

Let $\mathcal{H}$ be finite dimensional and let $\mathcal{N} \subset \mathcal{B}(\mathcal{H})$ be a nontrivial decoherence-free algebra, meaning

$$\mathcal{N} \neq \mathbb{C}I, \qquad \mathcal{N} \neq \mathcal{B}(\mathcal{H}).$$

Let $E_{\mathcal{N}} : \mathcal{B}(\mathcal{H}) \to \mathcal{N}$ be the σ_{tr}-preserving conditional expectation used in arXiv:1803.05379. For $p > 1$ and $q = p/(p-1)$, the p-Dirichlet form of a generator $\mathcal{L}$ is

$$\mathcal{E}_{p,\mathcal{L}}(X) = -\frac{p}{2(p-1)} \langle I_{q,p}(X), \mathcal{L}(X) \rangle_{\sigma_{\mathrm{tr}}},$$

where

$$I_{q,p}(X) = \Gamma_{\sigma_{\mathrm{tr}}}^{-1/q} \left(\left| \Gamma_{\sigma_{\mathrm{tr}}}^{1/p}(X) \right|^{p/q} \right).$$

The strong decoherence-free p-LSI is

$$\mathrm{Ent}_{p,\mathcal{N}}(X) \le c\,\mathcal{E}_{p,\mathcal{L}}(X) \qquad (X > 0).$$

We use the following identity from Bardet–Rouze, Lemma 3.5: for any density matrix ρ,

$$\mathrm{Ent}_{p,\mathcal{N}}\left(\Gamma_{\sigma_{\mathrm{tr}}}^{-1/p}(\rho^{1/p})\right) = \frac{1}{p} D(\rho \| E_{\mathcal{N}*}\rho).$$

Proof Intuition

The $p = 2$ no-go in the source paper is driven by a two-level perturbation. One eigenvalue inside the projected state $E_{\mathcal{N}*}\rho$ is made very small, while a tiny off-diagonal coherence is inserted between that small eigenspace and a fixed positive eigenspace. The relative entropy of the perturbation has a second variation containing

$$\frac{\log x - \log y}{x - y},$$

which diverges as $x \downarrow 0$. The Dirichlet energy, however, only sees divided differences of power functions. For every finite $p > 1$, the relevant powers $t^{1/p}$ and $t^{1/q}$ have bounded off-diagonal divided differences between $x \downarrow 0$ and $y > 0$. Thus entropy grows like $\log(1/x)$, while energy stays bounded.

Main Result

Theorem 1 (Strong p-LSI obstruction for nontrivial decoherence)**.** *Let $1 < p < \infty$. Let $(P_t)_{t\ge 0}$ be a finite-dimensional decohering quantum Markov semigroup with generator $\mathcal{L}$, decoherence-free algebra $\mathcal{N}$, and reference state σ_{tr}. Assume the decoherence is nontrivial:*

$$\mathcal{N} \ne \mathbb{C}I, \qquad \mathcal{N} \ne \mathcal{B}(\mathcal{H}).$$

Then no finite constant c satisfies

$$\mathrm{Ent}_{p,\mathcal{N}}(X) \le c\,\mathcal{E}_{p,\mathcal{L}}(X) \qquad (X > 0).$$

Consequently, among finite-dimensional nontrivially decohering QMS, the only possible strong endpoint in the family $\mathrm{LSI}_{p,\mathcal{N}}(c, 0)$ is $p = 1$.

Proof. We use the same two-level construction as the proof of Theorem 5.1 in Bardet–Rouze. Since $\mathcal{N}$ is nontrivial, there are density matrices $\rho_{\mathcal{N},k} \in E_{\mathcal{N}*}(\mathcal{D}(\mathcal{H}))$, unit vectors e_1, e_2, constants $\lambda_1, \lambda_2 > 0$, and a self-adjoint perturbation

$$\Delta = |e_1\rangle\langle e_2| + |e_2\rangle\langle e_1|$$

such that $E_{\mathcal{N}*}\Delta = 0$,

$$\rho_{k,\epsilon} := \rho_{\mathcal{N},k} + \epsilon\Delta$$

is a density matrix for sufficiently small $\epsilon > 0$, and e_1, e_2 are eigenvectors of $\rho_{\mathcal{N},k}$ with eigenvalues

$$x_k = \lambda_1/k, \qquad y_k = (1 - 1/k)\lambda_2.$$

Consequently $E_{\mathcal{N}*}\rho_{k,\epsilon} = \rho_{\mathcal{N},k}$.

Set

$$X_{k,\epsilon} := \Gamma_{\sigma_{\mathrm{tr}}}^{-1/p}(\rho_{k,\epsilon}^{1/p}).$$

The entropy identity recalled above and the standard second variation of relative entropy give, for fixed k and $\epsilon \downarrow 0$,

$$\mathrm{Ent}_{p,\mathcal{N}}(X_{k,\epsilon}) = \frac{\epsilon^2}{p}\, g(x_k, y_k)\, |\langle e_1, \Delta e_2 \rangle|^2 + O_k(\epsilon^3),$$

where

$$g(x,y) = \begin{cases} \dfrac{\log x - \log y}{x - y}, & x \neq y, \\ x^{-1}, & x = y. \end{cases}$$

Since $x_k \downarrow 0$ and $y_k \to \lambda_2 > 0$, we have

$$g(x_k, y_k) \longrightarrow +\infty.$$

It remains to show that the corresponding Dirichlet second variation is uniformly bounded in k. Put

$$\alpha = \frac{1}{p}, \qquad \beta = \frac{1}{q} = 1 - \frac{1}{p}.$$

Functional calculus for the two-dimensional off-diagonal perturbation gives

$$\rho_{k,\epsilon}^{\alpha} = \rho_{\mathcal{N},k}^{\alpha} + \epsilon a_k \Delta + O_k(\epsilon^2), \qquad \rho_{k,\epsilon}^{\beta} = \rho_{\mathcal{N},k}^{\beta} + \epsilon b_k \Delta + O_k(\epsilon^2),$$

where

$$a_k = \frac{x_k^{\alpha} - y_k^{\alpha}}{x_k - y_k}, \qquad b_k = \frac{x_k^{\beta} - y_k^{\beta}}{x_k - y_k}.$$

Because $0 < \alpha, \beta < 1$, both sequences (a_k) and (b_k) are bounded and converge to finite limits as $k \to \infty$.

The associated expansions are

$$X_{k,\epsilon} = X_{k,0} + \epsilon U_k + O_k(\epsilon^2), \qquad I_{q,p}(X_{k,\epsilon}) = Y_{k,0} + \epsilon V_k + O_k(\epsilon^2),$$

where $X_{k,0}, Y_{k,0} \in \mathcal{N}$, $U_k, V_k \in \ker E_{\mathcal{N}}$, and U_k, V_k are uniformly bounded in k. The first-order coefficient in the Dirichlet form vanishes. Indeed, contractivity of P_t on $L_p(\sigma_{\mathrm{tr}})$ gives non-negativity of $\mathcal{E}_{p,\mathcal{L}}$, and both signs of ϵ give admissible positive perturbations; hence $\epsilon = 0$ is a local minimum of the energy along this line.

For the second-order coefficient, only two kinds of terms can occur. The term

$$\langle V_k, \mathcal{L}(U_k) \rangle_{\sigma_{\mathrm{tr}}}$$

is uniformly bounded in k, since $\mathcal{L}$ is a fixed finite-dimensional operator and U_k, V_k are uniformly bounded. The remaining terms involve second-order corrections inside $\mathcal{N}$. On $\mathcal{N}$, the semigroup acts by a unitary automorphism group, so its generator is a derivation

$$\delta(A) = i[H, A] \qquad (A \in \mathcal{N})$$

for some self-adjoint $H \in \mathcal{N}$, after identifying the finite-dimensional representation. The state σ_{tr} is tracial on $\mathcal{N}$. Moreover, the $\mathcal{N}$-parts of the second-order corrections are obtained from the same two spectral projections of $\rho_{\mathcal{N},k}$, hence are functions of $X_{k,0}$ and commute with $Y_{k,0}$. Therefore, for every such correction $R_k \in \mathcal{N}$,

$$\langle Y_{k,0}, \delta(R_k)\rangle_{\sigma_{\mathrm{tr}}} = i \ \mathrm{tr}\big(\sigma_{\mathrm{tr}} Y_{k,0}[H, R_k]\big) = 0$$

by cyclicity of the tracial state on $\mathcal{N}$ and $[Y_{k,0}, R_k] = 0$. Hence the potentially singular fixed-algebra second-order terms do not contribute. Thus

$$\mathcal{E}_{p,\mathcal{L}}(X_{k,\epsilon}) = -\frac{p}{2(p-1)}\epsilon^2 \langle V_k, \mathcal{L}(U_k)\rangle_{\sigma_{\mathrm{tr}}} + O_k(\epsilon^3).$$

There exists $C_p < \infty$, independent of k, such that

$$\limsup_{\epsilon \downarrow 0} \frac{\mathcal{E}_{p,\mathcal{L}}(X_{k,\epsilon})}{\mathrm{Ent}_{p,\mathcal{N}}(X_{k,\epsilon})} \leq \frac{C_p}{g(x_k, y_k)}.$$

Letting $k \to \infty$ makes the right side vanish.

If a finite c satisfied the strong p-LSI, then the same quotient would be bounded below by $1/c$ for every k and sufficiently small ϵ, a contradiction. □

Remark 1 (Endpoint)**.** *Bardet's modified decoherence-free logarithmic Sobolev inequality is the* $p = 1$ *endpoint of this family, and positive modified-LSI constants are known in finite-dimensional examples under* L_1*-regularity assumptions. Thus this theorem rules out every finite* $p > 1$*, while the previously known modified-LSI theory supplies the endpoint cases.*

Verification Notes

The proof reduces exactly to the mechanism in the source paper's $p = 2$ obstruction, replacing the square-root divided difference by the pair of divided differences for $t^{1/p}$ and $t^{1/q}$. The first new point is that these divided differences remain bounded for every finite $p > 1$, while the relative-entropy divided difference diverges logarithmically. The second new point is the fixed-algebra cancellation, which removes the only second-order terms whose divided differences could be singular.

The script `code/diagonal_decoherence_ratio.py` checks the same asymptotic in the two-dimensional diagonal decoherence model, where the quotient is explicit.

Limitations and Novelty Check

The proof is finite dimensional and uses the standard structure theorem for decoherence-free algebras of finite-dimensional QMS. Infinite-dimensional extensions are not claimed.

Bounded literature search on 2026-06-23 used the phrases $\mathrm{LSI}_{p,\mathcal{N}}(c, 0)$, non-primitive quantum Markov semigroups, strong log-Sobolev, Bardet–Rouze, and amalgamated L_p. The search found the source arXiv record, Bardet's modified-LSI preprint, and later complete modified-LSI work, but no later explicit resolution of this strong p-range question.

References

Ivan Bardet and Cambyse Rouze. *Hypercontractivity and logarithmic Sobolev Inequality for non-primitive quantum Markov semigroups and estimation of decoherence rates.* arXiv:1803.05379, 2018.

Ivan Bardet. *Estimating the decoherence time using non-commutative Functional Inequalities.* arXiv:1710.01039, 2017.